

Basic Python Concepts

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1. Bus Seat Reservation:

Code:

```
seats = {"A1", "A2", "A3"}
seat = input("Enter seat number: ").strip().upper()
if seat in seats:
    print("Seat booked")
    seats.remove(seat)
else:
    print("Seat not available")
```

Output:

```
● PS C:\Users\Raju> python -u "d:\Projects\Python Projects\Extra\Practice.py"
Enter seat number: A1
Seat booked
● PS C:\Users\Raju> python -u "d:\Projects\Python Projects\Extra\Practice.py"
Enter seat number: A4
Seat not available
```

2. Simple Interest Calculator:

Code:

```
p = float(input("Principal: "))
r = float(input("Rate: "))
t = float(input("Time: "))
interest = (p * r * t) / 100
print("Simple Interest:", interest)
```

Output:

```
● PS C:\Users\Raju> python -u "d:\Projects\Python Projects\Extra\Practice.py"
Principal: 10
Rate: 5
Time: 6
Simple Interest: 3.0
```

3. Number Comparison:

Code:

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
if a > b:
    print("A is greater")
elif b > a:
    print("B is greater")
else:
    print("Both equal")
```

Output:

```
● PS C:\Users\Raju> python -u "d:\Projects\Python Projects\Extra\Practice.py"
Enter first number: 17
Enter second number: 23
B is greater
```

4. Marks Storage:

Code:

```
marks = []
for i in range(4):
    m = float(input("Enter marks: "))
    marks.append(m)
print("Highest marks:", max(marks))
```

Output:

```
● PS C:\Users\Raju> python -u "d:\Projects\Python Projects\Extra\tempCodeRunnerFile.py"
Enter marks: 92
Enter marks: 98
Enter marks: 96
Enter marks: 100
Highest marks: 100.0
```

5. True or False Quiz:

Code:

```
score = 0
if input("Sky is blue? ").strip().lower() == "true":
    score += 1
if input("10 < 5? ").strip().lower() == "false":
    score += 1
print("Score:", score)
```

Output:

```
PS C:\Users\Raju> python -u "d:\Projects\Python Projects\Extra\tempCodeRunnerFile.py"
Sky is blue? true
10 < 5? false
Score: 2
```

6. Random Number Generator:

Code:

```
import random
num = random.randint(1, 100)
print("Random number:", num)
```

Output:

```
PS C:\Users\Raju> python -u "d:\Projects\Python Projects\Extra\Practice.py"
Random number: 40
PS C:\Users\Raju> python -u "d:\Projects\Python Projects\Extra\Practice.py"
Random number: 36
```

7. PIN Verification:

Code:

```
pin = 5678
entry = 0
while entry != pin:
    try:
        entry = int(input("Enter PIN: "))
    except ValueError:
        print("Please enter numeric digits only.")
print("Access Granted")
```

Output:

```
● PS C:\Users\Raju> python -u "d:\Projects\Python Projects\Extra\Practice.py"
Enter PIN: abcd
Please enter numeric digits only.
Enter PIN: 8854
Enter PIN: 5678
Access Granted
```

8. Product Price Lookup:

Code:

```
products = {"Pen": 10, "Book": 50}
item = input("Enter product: ").strip().title()
print(products.get(item, "Not available"))
```

Output:

```
● PS C:\Users\Raju> python -u "d:\Projects\Python Projects\Extra\tempCodeRunnerFile.py"
Enter product: Book
50
● PS C:\Users\Raju> python -u "d:\Projects\Python Projects\Extra\tempCodeRunnerFile.py"
Enter product: Pencil
Not available
```

9. Square Finder:

Code:

```
def square(n):
    return n * n
num = float(input("Enter number: "))
print("Square:", square(num))
```

Output:

```
● PS C:\Users\Raju> python -u "d:\Projects\Python Projects\Extra\Practice.py"
Enter number: 5
Square: 25.0
```

10. Unique Visitors Counter:

Code:

```
visitors = set()
for i in range(3):
    name = input("Enter visitor name: ").strip().title()
    visitors.add(name)
print("Total unique visitors:", len(visitors))
```

Output:

```
● PS C:\Users\Raju> python -u "d:\Projects\Python Projects\Extra\tempCodeRunnerFile.py"
Enter visitor name: Aditya
Enter visitor name: Aman
Enter visitor name: Rohan
Total unique visitors: 3
```